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FRTE 1:1 Demographic Differentials Summary

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Ongoing Face Evaluations (<https://www.nist.gov/programs-projects/face-recognition-vendor-test-frvt-ongoing>)

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FRTE 1:N Identification (<https://pages.nist.gov/frvt/html/frvt1N.html>)

FRTE Demographic Effects (https://pages.nist.gov/frvt/html/frvt_demographics.html)

FRTE Face Mask Effects (https://pages.nist.gov/frvt/html/frvt_facemask.html)

FRTE Paperless Travel (https://pages.nist.gov/frvt/html/frvt_paperless_travel.html)

FRTE Twins Demonstration (https://pages.nist.gov/frvt/html/frvt_twins_demonstration.html)

FRTE FIVE (https://pages.nist.gov/frvt/html/frte_five.html)

FATE Projects

FATE MORPH (https://pages.nist.gov/frvt/html/frvt_morph.html)

FATE Quality (https://pages.nist.gov/frvt/html/frvt_quality.html)

FATE PAD (https://pages.nist.gov/frvt/html/frvt_pad.html)

FATE Age Estimation & Verification (https://pages.nist.gov/frvt/html/frvt_age_estimation.html)

Face Recognition Technology Evaluation: Demographic Effects in Face Recognition

Latest 1:1 Report (https://pages.nist.gov/frvt/reports/11/frvt_11_report.pdf) | Latest 1:N Report (https://pages.nist.gov/frvt/reports/1N/frvt_1N_report.pdf)

▼ Overview

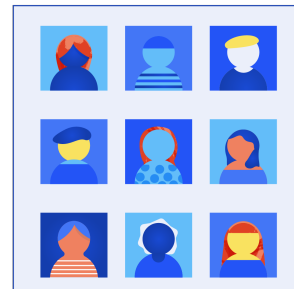
This page summarizes and links to all FRTE data and reports related to demographic effects in face recognition.

2022-07-12 NIST Interagency Report 8429: FRTE Part 8: Summarizing

Demographic Differentials PDF (https://pages.nist.gov/frvt/reports/demographics/nistir_8429.pdf)

2019-12-20 NIST Interagency Report 8280: FRTE Part 3: Demographics Effects PDF (https://pages.nist.gov/frvt/reports/demographics/nistir_8280.pdf)

The next section includes draft summary measures for demographic differentials for 1:1 recognition algorithms.



Credit: Natasha Hanacek/NIST

▼ FRTE 1:1 Demographic Differentials Summary

The table, last updated on 2025-03-05, includes summary indicators for how the two fundamental error rates vary by age, sex, and race. These are false negative demographic effects i.e. failures to associate two photos of an individual, and false positive demographic effects i.e. incorrect association of two photos of different individuals. **False negatives** are strongly dependent on image quality, and poor photography of a face can induce a demographic effect. Two examples that will elevate false negative rates: 1. inadequate lighting or under-exposure of dark-skinned individuals, or over-exposure of fair-skinned subjects; 2. failure to adjust a

camera for very tall or short individuals can lead to a pitch-angle variation **False positive** variations occur even with good image quality; they arise when an algorithm produces similarity score distributions that are displaced for one demographic versus another. This can occur due to under-representation of a demographic in the image dataset used for algorithm training.

False Positive Demographics	False Negative Demographics
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False Positive Differentials

The table below - last updated on 2025-03-05 - includes summary indicators for false positive demographic effects i.e. incorrect associations of two photos of two different individuals, the rate of which differs by age, sex or race. False positives occur primarily due to anatomic similarity of the faces and this occurs in images of pristine quality. False positives can in principle occur also due to image artifacts, such as reflections or eye-glasses, but these are specific to particular recognition algorithms. FMR values are measured over comparisons of two high quality frontal portrait images of two people of the same sex, same age group, and same region of birth. The threshold is fixed for each algorithm to give FMR of 0.00003 overall.

- The rows list algorithms submitted to the 1:1 track of FRTE. The relative importance of demographic effects in FNMR versus FMR is highly application dependent. The columns are:
- 1: Algorithm name
 - 2: Date algorithm was submitted to NIST
 - 3: A summary FNMR value (so that readers can look at the more accurate algorithms first). Smaller values are better.
 - 4: The best FMR value for any one demographic - this is the lowest across 5 regions, 5 age groups and two sexes
 - 5: The worst FMR value for any one demographic - this is the highest across 5 regions, 5 age groups and two sexes
 - 6: The ratio of worst to best FMR given in prior two columns i.e. how-many times larger one is than the other. Smaller values are better
 - 7: The ratio of worst FMR to arithmetic mean of FMR of all demographics
 - 8: The ratio of worst FMR to geometric mean of FMR of all demographics
 - 9: The arithmetic mean of absolute value of log10 of all ratios FMR / X where X = geometric mean of FMR, over all demographics. This measures the spread of FMR values around their geometric mean. Smaller values are better
 - 10: The Gini coefficient for FMR, the mean absolute difference across all groups, divided by the mean
 - 11: FMR E. Africa over FMR E. Europe - numerator and denominator are geometric means of 10 FMR ratios (5 age groups, 2 sexes). Ratios close to 1 are best, and ranked accordingly
 - 12: FMR E. Asia over FMR E. Europe - numerator and denominator are geometric means of 10 FMR ratios (5 age groups, 2 sexes). Ratios close to 1 are best, and ranked accordingly
 - 13: FMR S. Asia over FMR E. Europe - numerator and denominator are geometric means of 10 FMR ratios (5 age groups, 2 sexes). Ratios close to 1 are best, and ranked accordingly
 - 14: FMR Female over FMR Male - numerator and denominator are geometric means of 25 FMR ratios estimates (5 regions, 5 age groups). Ratios close to 1 are best, and ranked accordingly
 - 15: FMR 65+ over FMR (20-35] - numerator and denominator are geometric means of 10 FMR ratios estimates (5 regions, 2 sexes). Ratios close to 1 are best, and ranked accordingly

A note on 1:N While this table lists results for 1:1 algorithms, it will have relevance to that subset of 1:N algorithms that implement 1:N search as N 1:1 comparisons followed by a sort operation. The demographic effects noted here will be material in 1:N operations and will be magnified if the gallery and the search stream include the affected demographic. This is discussed in the Annex, and in the academic literature PDF (https://www.dhs.gov/sites/default/files/publications/21_0922_st_quantifying-commercial-face-recognition-gender-and-race_updated.pdf), PDF (https://openaccess.thecvf.com/content/ICCV2021W/RPRMI/papers/Drozdzowski_The_Watchlist_Imbalance_Effect_in_Biometric_Face_Identification_Comparing_Theoretical_ICCVW_2021_paper.pdf). Note those publications use the Binomial model of 1:N search and it is known that some 1:N algorithms do not implement 1:N simply as N 1:1 comparisons, so the model does not apply. Demographic effects there must be measured in separate empirical 1:N tests.

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Algorithm	Submission Date	FNMR Overall	FMR Min	FMR Max	FMR Max/Min	FMR Max/Mean	FMR Max/GeoMean	
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qazsmartvisionai_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/qazsmartvisionai_003.pdf)	2025-01-30	0.0013 ⁽¹⁾	0.00006 E.Europe M (35-50]	0.02150 W.Africa F (65-99]	345 ⁽¹⁸⁵⁾	13.30 ⁽⁴⁸⁴⁾
qazsmartvisionai_002 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/qazsmartvisionai_002.pdf)	2024-09-25	0.0013 ⁽²⁾	0.00009 E.Europe M (35-50]	0.02020 W.Africa F (65-99]	215 ⁽⁷⁷⁾	12.63 ⁽⁴⁴⁸⁾
recognito_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/recognito_001.pdf)	2023-09-27	0.0013 ⁽³⁾	0.00001 E.Europe M (20-35]	0.01878 W.Africa F (65-99]	3304 ⁽⁵⁷¹⁾	15.28 ⁽⁵³⁸⁾
qazsmartvisionai_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/qazsmartvisionai_001.pdf)	2024-05-21	0.0013 ⁽⁴⁾	0.00017 E.Europe M (20-35]	0.02402 W.Africa F (65-99]	139 ⁽³⁵⁾	11.26 ⁽³¹²⁾
recognito_000 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/recognito_000.pdf)	2023-05-24	0.0014 ⁽⁵⁾	0.00001 E.Europe M (20-35]	0.02113 W.Africa F (65-99]	3897 ⁽⁵⁷³⁾	15.89 ⁽⁵⁵⁶⁾
stcon_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/stcon_005.pdf)	2024-11-13	0.0014 ⁽⁶⁾	0.00001 E.Europe M (35-50]	0.00632 W.Africa F (65-99]	475 ⁽³⁸¹⁾	10.85 ⁽²⁶⁷⁾
qazsmartvisionai_000 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/qazsmartvisionai_000.pdf)	2024-01-11	0.0015 ⁽⁷⁾	0.00008 E.Europe M (35-50]	0.01282 W.Africa F (65-99]	155 ⁽⁵¹⁾	9.29 ⁽¹²⁵⁾
viante_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/viante_001.pdf)	2024-12-30	0.0015 ⁽⁸⁾	0.00000 E.Europe F (20-35]	0.02794 W.Africa F (65-99]	87832 ⁽⁵⁹⁴⁾	21.48 ⁽⁵⁹¹⁾
sensetime_010 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/sensetime_010.pdf)	2024-12-03	0.0015 ⁽⁹⁾	0.00004 E.Europe M (35-50]	0.01947 W.Africa F (65-99]	493 ⁽⁴⁰¹⁾	15.35 ⁽⁵⁴¹⁾
cloudwalk_mt_007 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/cloudwalk_mt_007.pdf)	2023-02-21	0.0015 ⁽¹⁰⁾	0.00012 E.Europe M (20-35]	0.00710 W.Africa F (65-99]	60 ⁽⁹⁾	8.22 ⁽⁶⁴⁾
roc_018 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/roc_018.pdf)	2025-01-31	0.0015 ⁽¹¹⁾	0.00005 E.Europe M (20-35]	0.01061 W.Africa F (65-99]	212 ⁽⁷⁶⁾	10.89 ⁽²⁷⁰⁾
cloudwalk_mt_006 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/cloudwalk_mt_006.pdf)	2022-10-20	0.0015 ⁽¹²⁾	0.00014 E.Europe M (20-35]	0.00917 W.Africa F (65-99]	65 ⁽¹⁴⁾	8.58 ⁽⁷⁹⁾
cloudwalk_mt_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/cloudwalk_mt_005.pdf)	2022-03-29	0.0015 ⁽¹³⁾	0.00013 E.Europe M (20-35]	0.01997 W.Africa F (65-99]	150 ⁽⁴⁷⁾	10.08 ⁽²⁰²⁾
sensetime_007 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/sensetime_007.pdf)	2022-06-17	0.0015 ⁽¹⁴⁾	0.00004 E.Europe M (20-35]	0.01565 W.Africa F (65-99]	402 ⁽²⁷⁰⁾	13.84 ⁽⁵⁰⁴⁾
intema_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/intema_001.pdf)	2023-01-11	0.0015 ⁽¹⁵⁾	0.00005 E.Europe M (20-35]	0.02071 W.Africa F (65-99]	399 ⁽²⁶¹⁾	13.12 ⁽⁴⁷⁸⁾

toshiba_010 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/toshiba_010.pdf)	2025-02-03	0.0016 ⁽¹⁶⁾	0.00002 E.Europe M (35-50]	0.00449 W.Africa F (65-99]	288 ⁽¹³³⁾	9.33 ⁽¹³²⁾
stcon_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/stcon_004.pdf)	2024-06-18	0.0016 ⁽¹⁷⁾	0.00001 E.Europe M (35-50]	0.00546 W.Africa F (65-99]	440 ⁽³⁴⁰⁾	10.27 ⁽²²²⁾
viante_000 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/viante_000.pdf)	2023-06-09	0.0016 ⁽¹⁸⁾	0.00000 E.Europe F (20-35]	0.01366 W.Africa F (65-99]	48367 ⁽⁵⁹³⁾	15.61 ⁽⁵⁴⁸⁾
deepsense_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/deepsense_005.pdf)	2024-10-28	0.0016 ⁽¹⁹⁾	0.00009 E.Europe M (20-35]	0.01330 W.Africa F (65-99]	149 ⁽⁴⁶⁾	10.77 ⁽²⁵⁹⁾
idemia_011 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/idemia_011.pdf)	2024-08-06	0.0016 ⁽²⁰⁾	0.00022 C.America M (12-20]	0.00118 W.Africa F (65-99]	5 ⁽¹⁾	2.35 ⁽¹⁾
vinbigdata_006 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/vinbigdata_006.pdf)	2025-02-19	0.0016 ⁽²¹⁾	0.00050 E.Europe M (35-50]	0.02687 W.Africa F (65-99]	54 ⁽⁷⁾	8.46 ⁽⁷³⁾
toshiba_009 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/toshiba_009.pdf)	2024-09-16	0.0016 ⁽²²⁾	0.00002 E.Europe M (35-50]	0.00429 W.Africa F (65-99]	241 ⁽¹⁰³⁾	8.96 ⁽¹⁰⁹⁾
sensetime_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/sensetime_008.pdf)	2023-01-04	0.0017 ⁽²³⁾	0.00005 E.Europe M (35-50]	0.01709 W.Africa F (65-99]	327 ⁽¹⁶⁶⁾	16.48 ⁽⁵⁷⁰⁾
cmcuni_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/cmcuni_001.pdf)	2023-12-05	0.0017 ⁽²⁴⁾	0.00003 E.Europe M (35-50]	0.01381 W.Africa F (65-99]	438 ⁽³³⁵⁾	12.01 ⁽³⁸⁵⁾
surrey_cvssp_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/surrey_cvssp_003.pdf)	2023-09-13	0.0017 ⁽²⁵⁾	0.00003 E.Europe M (35-50]	0.00826 W.Africa F (65-99]	303 ⁽¹⁴³⁾	10.26 ⁽²²⁰⁾
idemia_010 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/idemia_010.pdf)	2023-06-30	0.0017 ⁽²⁶⁾	0.00023 E.Europe M (35-50]	0.01449 W.Africa F (65-99]	64 ⁽¹²⁾	7.42 ⁽³⁴⁾
psl_012 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/psl_012.pdf)	2023-10-13	0.0017 ⁽²⁷⁾	0.00002 E.Europe M (35-50]	0.00802 W.Africa F (65-99]	527 ⁽⁴²⁶⁾	10.98 ⁽²⁷⁷⁾
toshiba_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/toshiba_008.pdf)	2024-02-01	0.0017 ⁽²⁸⁾	0.00002 E.Europe M (35-50]	0.00439 W.Africa F (65-99]	260 ⁽¹¹⁵⁾	9.32 ⁽¹³¹⁾
psl_014 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/psl_014.pdf)	2024-09-11	0.0017 ⁽²⁹⁾	0.00002 E.Europe M (35-50]	0.00802 W.Africa F (65-99]	527 ⁽⁴²⁶⁾	10.98 ⁽²⁷⁶⁾
psl_013 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/psl_013.pdf)	2024-04-18	0.0017 ⁽³⁰⁾	0.00002 E.Europe M (35-50]	0.00802 W.Africa F (65-99]	527 ⁽⁴²⁶⁾	10.98 ⁽²⁷⁵⁾
neurotechnology_019 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/neurotechnology_019.pdf)	2024-07-31	0.0017 ⁽³¹⁾	0.00007 E.Europe M (35-50]	0.02435 W.Africa F (65-99]	326 ⁽¹⁶⁴⁾	12.71 ⁽⁴⁵⁵⁾

useb_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_useb_004.pdf)	2024-12-23	0.0017 ⁽³²⁾	0.00001 E.Europe M (35-50]	0.00441 W.Africa F (65-99]	709 ⁽⁵⁰⁰⁾	13.56 ⁽⁴⁹³⁾
stcon_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_stcon_003.pdf)	2023-11-07	0.0017 ⁽³³⁾	0.00001 E.Europe M (35-50]	0.00508 W.Africa F (65-99]	783 ⁽⁵¹⁴⁾	12.65 ⁽⁴⁴⁹⁾
alchera_009 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_alchera_009.pdf)	2024-11-14	0.0017 ⁽³⁴⁾	0.00001 E.Europe M (35-50]	0.00441 W.Africa F (65-99]	709 ⁽⁵⁰⁰⁾	13.56 ⁽⁴⁹²⁾
innovatrics_013 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_innovatrics_013.pdf)	2024-11-13	0.0017 ⁽³⁵⁾	0.00003 E.Europe M (35-50]	0.01304 W.Africa F (65-99]	389 ⁽²⁴⁸⁾	11.15 ⁽²⁹⁸⁾
roc_017 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_roc_017.pdf)	2024-06-24	0.0017 ⁽³⁶⁾	0.00006 E.Europe M (20-35]	0.00643 W.Africa F (65-99]	104 ⁽¹⁹⁾	9.49 ⁽¹⁵²⁾
deepsense_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_deepsense_004.pdf)	2024-06-21	0.0017 ⁽³⁷⁾	0.00008 E.Europe M (20-35]	0.01154 W.Africa F (65-99]	152 ⁽⁴⁸⁾	11.11 ⁽²⁹⁰⁾
paravision_017 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_paravision_017.pdf)	2025-01-13	0.0017 ⁽³⁸⁾	0.00000 S.Asia M (35-50]	0.00139 E.Europe M (12-20]	541 ⁽⁴⁴¹⁾	8.37 ⁽⁶⁷⁾
cybercore_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_cybercore_003.pdf)	2022-08-31	0.0017 ⁽³⁹⁾	0.00003 E.Europe M (35-50]	0.00947 W.Africa F (65-99]	338 ⁽¹⁷⁵⁾	11.09 ⁽²⁸⁵⁾
papil11_002 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_papil11_002.pdf)	2025-01-02	0.0017 ⁽⁴⁰⁾	0.00004 E.Europe M (20-35]	0.00547 W.Africa F (65-99]	143 ⁽³⁷⁾	7.78 ⁽⁴⁶⁾
papil11_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_papil11_001.pdf)	2024-08-06	0.0018 ⁽⁴¹⁾	0.00004 E.Europe M (20-35]	0.00548 W.Africa F (65-99]	146 ⁽⁴¹⁾	7.78 ⁽⁴⁵⁾
roc_014 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_roc_014.pdf)	2022-12-21	0.0018 ⁽⁴²⁾	0.00008 E.Asia M (20-35]	0.01871 W.Africa F (65-99]	236 ⁽⁹⁹⁾	13.47 ⁽⁴⁹¹⁾
kakao_009 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_kakao_009.pdf)	2023-11-07	0.0018 ⁽⁴³⁾	0.00002 E.Europe M (35-50]	0.00623 W.Africa F (65-99]	372 ⁽²³⁰⁾	14.20 ⁽⁵¹⁸⁾
megvii_009 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_megvii_009.pdf)	2024-02-12	0.0018 ⁽⁴⁴⁾	0.00006 E.Asia M (20-35]	0.01441 W.Africa F (65-99]	227 ⁽⁹²⁾	12.51 ⁽⁴³⁶⁾
kakao_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_kakao_008.pdf)	2022-05-12	0.0018 ⁽⁴⁵⁾	0.00003 E.Europe M (35-50]	0.00730 W.Africa F (65-99]	282 ⁽¹²⁷⁾	11.35 ⁽³²⁵⁾
megvii_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_megvii_005.pdf)	2022-03-28	0.0018 ⁽⁴⁶⁾	0.00001 E.Asia M (20-35]	0.01059 W.Africa F (65-99]	1608 ⁽⁵⁵⁹⁾	21.05 ⁽⁵⁹⁰⁾
toshiba_007 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_toshiba_007.pdf)	2023-09-13	0.0018 ⁽⁴⁷⁾	0.00001 E.Europe M (35-50]	0.00850 W.Africa F (65-99]	599 ⁽⁴⁷¹⁾	12.77 ⁽⁴⁶¹⁾

sensetime_006 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/sensetime_006.pdf)	2021-12-28	0.0018 ⁽⁴⁸⁾	0.00005 E.Europe M (20-35]	0.02376 W.Africa F (65-99]	442 ⁽³⁴³⁾	14.03 ⁽⁵¹⁴⁾
samsungsd_002 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/samsungsd_002.pdf)	2022-09-16	0.0018 ⁽⁴⁹⁾	0.00002 E.Europe M (35-50]	0.00797 W.Africa F (65-99]	426 ⁽³²⁰⁾	11.40 ⁽³³²⁾
adera_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/adera_004.pdf)	2022-11-14	0.0018 ⁽⁵⁰⁾	0.00001 E.Europe F (20-35]	0.02001 W.Africa F (65-99]	1925 ⁽⁵⁶³⁾	24.52 ⁽⁵⁹⁴⁾
vinbigdata_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/vinbigdata_005.pdf)	2024-10-16	0.0018 ⁽⁵¹⁾	0.00009 E.Europe M (35-50]	0.00942 W.Africa F (65-99]	111 ⁽²¹⁾	9.51 ⁽¹⁵⁴⁾
tnitech_000 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/tnitech_000.pdf)	2024-02-05	0.0018 ⁽⁵²⁾	0.00003 E.Europe M (35-50]	0.01023 W.Africa F (65-99]	364 ⁽²¹⁹⁾	11.27 ⁽³¹⁴⁾
psl_011 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/psl_011.pdf)	2022-10-06	0.0018 ⁽⁵³⁾	0.00001 E.Europe M (50-65]	0.00587 W.Africa F (65-99]	887 ⁽⁵²⁶⁾	10.28 ⁽²²³⁾
frpkauai_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/frpkauai_003.pdf)	2024-05-30	0.0018 ⁽⁵⁴⁾	0.00003 E.Europe M (35-50]	0.00896 W.Africa F (65-99]	305 ⁽¹⁴⁶⁾	10.66 ⁽²⁵⁰⁾
surrey_cvssp_002 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/surrey_cvssp_002.pdf)	2023-02-16	0.0018 ⁽⁵⁵⁾	0.00003 E.Europe M (35-50]	0.00829 W.Africa F (65-99]	305 ⁽¹⁴⁴⁾	10.24 ⁽²¹⁸⁾
roc_016 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/roc_016.pdf)	2023-12-19	0.0018 ⁽⁵⁶⁾	0.00007 E.Europe F (12-20]	0.00831 W.Africa F (65-99]	122 ⁽³⁰⁾	8.72 ⁽⁸⁸⁾
rebs_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/rebs_001.pdf)	2023-12-22	0.0018 ⁽⁵⁷⁾	0.00000 E.Europe M (20-35]	0.00486 W.Africa F (65-99]	1505 ⁽⁵⁵⁶⁾	12.15 ⁽⁴⁰⁵⁾
paravision_015 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/paravision_015.pdf)	2024-09-09	0.0019 ⁽⁵⁸⁾	0.00000 S.Asia M (35-50]	0.00126 E.Europe M (12-20]	734 ⁽⁵⁰⁵⁾	8.81 ⁽⁹⁴⁾
clearviewai_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/clearviewai_001.pdf)	2023-12-05	0.0019 ⁽⁵⁹⁾	0.00002 E.Europe M (20-35]	0.00845 W.Africa F (65-99]	425 ⁽³¹⁹⁾	14.64 ⁽⁵²⁶⁾
neurotechnology_018 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/neurotechnology_018.pdf)	2024-02-09	0.0019 ⁽⁶⁰⁾	0.00005 E.Europe M (35-50]	0.01089 W.Africa F (65-99]	238 ⁽¹⁰²⁾	10.22 ⁽²¹⁶⁾
paravision_013 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/paravision_013.pdf)	2023-05-08	0.0019 ⁽⁶¹⁾	0.00000 S.Asia M (35-50]	0.00086 E.Europe M (12-20]	458 ⁽³⁶⁶⁾	5.90 ⁽¹¹⁾
firstcreditkz_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/firstcreditkz_004.pdf)	2024-10-02	0.0019 ⁽⁶²⁾	0.00003 E.Europe M (35-50]	0.00854 W.Africa F (65-99]	258 ⁽¹¹³⁾	9.91 ⁽¹⁸⁶⁾
nhn_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/nhn_005.pdf)	2023-12-06	0.0019 ⁽⁶³⁾	0.00002 E.Europe M (35-50]	0.00951 W.Africa F (65-99]	539 ⁽⁴³⁷⁾	11.66 ⁽³⁵⁵⁾

nominder_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/nominder_004.pdf)	2024-07-11	0.0019 ⁽⁶⁴⁾	0.00002 E.Europe M (35-50]	0.00970 W.Africa F (65-99]	427 ⁽³²²⁾	11.36 ⁽³²⁶⁾
roc_Q15 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/roc_Q15.pdf)	2023-07-20	0.0019 ⁽⁶⁵⁾	0.00004 S.Asia M (20-35]	0.00650 W.Africa F (65-99]	167 ⁽⁵⁵⁾	10.28 ⁽²²⁸⁾
pyramid_Q01 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/pyramid_Q01.pdf)	2024-11-26	0.0019 ⁽⁶⁶⁾	0.00001 E.Europe M (35-50]	0.00486 W.Africa F (65-99]	601 ⁽⁴⁷³⁾	13.78 ⁽⁵⁰²⁾
omnigarde_Q04 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/omnigarde_Q04.pdf)	2024-06-24	0.0019 ⁽⁶⁷⁾	0.00004 E.Europe M (35-50]	0.00915 W.Africa F (65-99]	248 ⁽¹⁰⁶⁾	11.03 ⁽²⁷⁹⁾
vocord_Q10 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/vocord_Q10.pdf)	2021-12-20	0.0019 ⁽⁶⁸⁾	0.00005 E.Europe M (20-35]	0.01232 W.Africa F (65-99]	272 ⁽¹²¹⁾	10.20 ⁽²¹²⁾
s1_Q05 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/s1_Q05.pdf)	2022-06-17	0.0019 ⁽⁶⁹⁾	0.00002 E.Europe M (35-50]	0.01039 W.Africa F (65-99]	519 ⁽⁴²⁰⁾	12.82 ⁽⁴⁶⁶⁾
nominder_Q05 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/nominder_Q05.pdf)	2024-11-12	0.0019 ⁽⁷⁰⁾	0.00002 E.Europe M (35-50]	0.00998 W.Africa F (65-99]	423 ⁽³¹³⁾	11.39 ⁽³³⁰⁾
datech_Q02 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/datech_Q02.pdf)	2024-09-11	0.0019 ⁽⁷¹⁾	0.00003 E.Europe M (35-50]	0.01008 W.Africa F (65-99]	339 ⁽¹⁷⁶⁾	12.27 ⁽⁴¹⁷⁾
clearviewai_Q02 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/clearviewai_Q02.pdf)	2024-05-22	0.0019 ⁽⁷²⁾	0.00002 E.Europe M (20-35]	0.01021 W.Africa F (65-99]	470 ⁽³⁷⁴⁾	14.39 ⁽⁵²²⁾
sensetime_Q05 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/sensetime_Q05.pdf)	2021-05-24	0.0019 ⁽⁷³⁾	0.00003 E.Europe M (35-50]	0.02442 W.Africa F (65-99]	790 ⁽⁵¹⁵⁾	15.99 ⁽⁵⁶²⁾
seventhsense_Q05 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/seventhsense_Q05.pdf)	2023-10-05	0.0019 ⁽⁷⁴⁾	0.00001 E.Europe M (35-50]	0.00437 W.Africa F (65-99]	635 ⁽⁴⁸⁵⁾	12.07 ⁽³⁹¹⁾
samsungds_Q01 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/samsungds_Q01.pdf)	2022-04-18	0.0019 ⁽⁷⁵⁾	0.00003 E.Europe M (35-50]	0.00738 W.Africa F (65-99]	218 ⁽⁸⁰⁾	9.22 ⁽¹²³⁾
maxvision_Q06 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/maxvision_Q06.pdf)	2024-01-19	0.0019 ⁽⁷⁶⁾	0.00002 E.Europe M (35-50]	0.00878 W.Africa F (65-99]	360 ⁽²¹⁵⁾	12.55 ⁽⁴³⁹⁾
cubox_Q03 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/cubox_Q03.pdf)	2023-03-07	0.0020 ⁽⁷⁷⁾	0.00001 E.Europe M (35-50]	0.00511 W.Africa F (65-99]	896 ⁽⁵²⁷⁾	13.74 ⁽⁴⁹⁹⁾
datech_Q01 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/datech_Q01.pdf)	2024-02-21	0.0020 ⁽⁷⁸⁾	0.00003 E.Europe M (35-50]	0.01012 W.Africa F (65-99]	347 ⁽¹⁹⁰⁾	12.18 ⁽⁴⁰⁸⁾
neurotechnology_Q17 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/neurotechnology_Q17.pdf)	2023-07-28	0.0020 ⁽⁷⁹⁾	0.00005 E.Europe M (35-50]	0.01104 W.Africa F (65-99]	236 ⁽¹⁰⁰⁾	10.20 ⁽²¹³⁾

cubox_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/cubox_004.pdf)	2024-12-05	0.0020 ⁽⁸⁰⁾	0.00002 E.Europe M (35-50]	0.00580 W.Africa F (65-99]	286 ⁽¹³²⁾	9.36 ⁽¹³⁵⁾
s1_007 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/s1_007.pdf)	2023-03-20	0.0020 ⁽⁸¹⁾	0.00003 E.Europe M (35-50]	0.00908 W.Africa F (65-99]	341 ⁽¹⁸¹⁾	11.94 ⁽³⁷⁷⁾
neurotechnology_016 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/neurotechnology_016.pdf)	2023-02-01	0.0020 ⁽⁸²⁾	0.00003 E.Europe M (35-50]	0.00933 W.Africa F (65-99]	313 ⁽¹⁵³⁾	11.27 ⁽³¹³⁾
nominder_000 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/nominder_000.pdf)	2023-02-06	0.0020 ⁽⁸³⁾	0.00002 E.Europe M (35-50]	0.00969 W.Africa F (65-99]	396 ⁽²⁵⁶⁾	11.28 ⁽³¹⁵⁾
vnpt_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/vnpt_008.pdf)	2024-09-20	0.0020 ⁽⁸⁴⁾	0.00001 E.Europe M (35-50]	0.00521 W.Africa F (65-99]	444 ⁽³⁴⁶⁾	11.17 ⁽³⁰¹⁾
maxvision_005 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/maxvision_005.pdf)	2023-09-21	0.0020 ⁽⁸⁵⁾	0.00002 E.Europe M (35-50]	0.00873 W.Africa F (65-99]	359 ⁽²¹²⁾	12.48 ⁽⁴³⁴⁾
chtface_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/chtface_008.pdf)	2024-09-27	0.0020 ⁽⁸⁶⁾	0.00001 E.Europe M (35-50]	0.00586 W.Africa F (65-99]	392 ⁽²⁵¹⁾	10.95 ⁽²⁷³⁾
idemia_009 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/idemia_009.pdf)	2022-07-27	0.0020 ⁽⁸⁷⁾	0.00027 C.America M (50-65]	0.00641 W.Africa F (65-99]	24 ⁽²⁾	7.01 ⁽²⁴⁾
vnpt_007 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/vnpt_007.pdf)	2024-04-29	0.0020 ⁽⁸⁸⁾	0.00001 E.Europe M (35-50]	0.00558 W.Africa F (65-99]	486 ⁽³⁹⁵⁾	11.91 ⁽³⁷⁵⁾
yuan_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/yuan_008.pdf)	2023-11-06	0.0020 ⁽⁸⁹⁾	0.00002 E.Europe M (35-50]	0.00992 W.Africa F (65-99]	566 ⁽⁴⁵⁶⁾	12.66 ⁽⁴⁵⁰⁾
surrey_cvssp_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/surrey_cvssp_004.pdf)	2024-07-11	0.0020 ⁽⁹⁰⁾	0.00003 E.Europe M (35-50]	0.00826 W.Africa F (65-99]	303 ⁽¹⁴²⁾	10.23 ⁽²¹⁷⁾
tech5_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/tech5_008.pdf)	2023-08-11	0.0021 ⁽⁹¹⁾	0.00002 E.Europe M (35-50]	0.00855 W.Africa F (65-99]	401 ⁽²⁶⁵⁾	11.21 ⁽³⁰⁴⁾
deepsense_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/deepsense_003.pdf)	2023-10-23	0.0021 ⁽⁹²⁾	0.00004 E.Europe M (35-50]	0.00558 W.Africa F (65-99]	130 ⁽³²⁾	9.71 ⁽¹⁷¹⁾
toshiba_006 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/toshiba_006.pdf)	2022-06-29	0.0021 ⁽⁹³⁾	0.00002 E.Europe M (35-50]	0.00866 W.Africa F (65-99]	376 ⁽²³³⁾	9.62 ⁽¹⁶⁵⁾
alchera_008 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/alchera_008.pdf)	2024-06-17	0.0021 ⁽⁹⁴⁾	0.00001 E.Europe M (35-50]	0.00537 W.Africa F (65-99]	421 ⁽³⁰⁷⁾	12.53 ⁽⁴³⁸⁾
inspur_001 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic/inspur_001.pdf)	2023-02-24	0.0021 ⁽⁹⁵⁾	0.00003 E.Europe M (35-50]	0.01051 W.Africa F (65-99]	401 ⁽²⁶⁶⁾	11.68 ⁽³⁵⁷⁾

cist_004 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_cist_004.pdf)	2024-02-15	0.0021 ⁽⁹⁶⁾	0.00002 E.Europe M (35-50]	0.01007 W.Africa F (65-99]	652 ⁽⁴⁸⁹⁾	11.52 ⁽³⁴⁵⁾		
useb_003 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_useb_003.pdf)	2024-08-09	0.0021 ⁽⁹⁷⁾	0.00001 E.Europe M (35-50]	0.00547 W.Africa F (65-99]	481 ⁽³⁸⁹⁾	12.71 ⁽⁴⁵⁴⁾		
alchera_006 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_alchera_006.pdf)	2023-07-18	0.0021 ⁽⁹⁸⁾	0.00001 E.Europe M (35-50]	0.00557 W.Africa F (65-99]	522 ⁽⁴²²⁾	12.84 ⁽⁴⁶⁹⁾		
useb_002 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_useb_002.pdf)	2024-01-19	0.0021 ⁽⁹⁹⁾	0.00001 E.Europe M (35-50]	0.00557 W.Africa F (65-99]	520 ⁽⁴²¹⁾	12.84 ⁽⁴⁶⁸⁾		
alchera_007 (https://pages.nist.gov/frvt/reports/demographics/figures/11/heatmap_fmr_demographic_x_demographic_alchera_007.pdf)	2023-11-22	0.0021 ⁽¹⁰⁰⁾	0.00001 E.Europe M (35-50]	0.00630 W.Africa F (65-99]	446 ⁽³⁴⁹⁾	11.49 ⁽³³⁸⁾		
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